

Total No. of Questions : 8]

**PD4638**

SEAT No. :

[Total No. of Pages : 1

**[6404]-144**

**B.E.(E & TC)**

**NANO ELECTRONICS**

**(2019 Pattern) (Semester - VIII) (Elective-VI) (404192 B)**

*Time : 2½ Hours]*

*[Max. Marks : 70*

*Instructions to the candidates:*

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

**Q1) a)** Explain Structure of CNT & Carbon nano tubes. **[8]**

**b)** Explain Nano material and its types. **[8]**

OR

**Q2) a)** Explain properties of CNT. **[8]**

**b)** Explain Method to Stabilize Nano particles. **[8]**

**Q3) a)** Explain Optical lithography process in detail. **[9]**

**b)** Explain Electron Beam Lithography with neat Diagram. **[9]**

OR

**Q4) a)** Explain Nano electronics for communication. **[9]**

**b)** Explain Atomic Lithography with neat diagram. **[9]**

**Q5) a)** What are molecular switch? Explain Ph switch. **[9]**

**b)** Explain MEMS. **[9]**

OR

**Q6) a)** Explain Nano tubes Actuators. **[9]**

**b)** Explain types of Super molecular Switches. **[9]**

**Q7) a)** Explain Piezo-Electric Biosensor. **[9]**

**b)** Which are types of Nano Sensor? Explain Nano biosensor. **[9]**

OR

**Q8) a)** What is use of Nano technology in Electronics? **[9]**

**b)** Explain Transformation. **[9]**

